Visvesvaraya Technological University, Belagavi

Jnana Sangama, Belagavi – 590 018

INTERNSHIP REPORT (BINT803B) ON

"ANDROID APP DEVELOPMENT USING GEN AI"

Internship Report submitted in partial fulfilment of the requirements for the Award of Degree of

BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE AND ENGINEERING

Submitted by
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1KS22CS113

Under the Guidance of
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Dept. of CSE, KSIT

Department of Computer Science & Engineering

K. S. INSTITUTE OF TECHNOLOGY

An Autonomous Institution under VTU, Approved by AICTE
Accredited by NBA (CSE & ECE), NAAC with A+
No.14, Raghuvanahalli, Kanakapura Road, Bengaluru - 560109
2025-2026

K. S. Institute of Technology

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Department of Computer Science & Engineering

CERTIFICATE

Certified that the **Internship Report (BINT803B)** entitled **"ANDROID APP DEVELOPMENT USING GEN AI"** is a bonafide work carried out by **RAKSHITA A U (1KS22CS113)** in fulfilment for VIII semester B.E., Internship in the branch of Computer Science and Engineering prescribed by **Visvesvaraya Technological University, Belagavi** during the period of February 2026 to May 2026. It is certified that all the corrections and suggestions indicated for internal assessment have been incorporated. The Internship Report has been approved as it satisfies the academic requirements in report of Internship prescribed for the Bachelor of Engineering degree.

…………………………	..…………….………	…..…..……………………
Signature of the Guide	**Signature of the HOD**	**Signature of the Principal & Director**
[Ms. Beena K]	[Dr. Rekha B. Venkatapura]	[Dr. Dilip Kumar K]]

CERTIFICATE

“Certified that this Internship report is an original report of work done by me under the guidance of Internship Mentor **Ms. Beena K** and under the Supervision of Internship Supervisor(Industry) **Mr. Tirumal Mutalik Desai** submitted as a part of the Internship Course of Undergraduate Programme of Visvesvaraya Technological University, Jnana Sangama, Machhe, Belagavi-590018”.

Date:

Signature of the student

Signature of the College Internship Supervisor

Signature of the Industry Internship Supervisor

Signature of the H.O.D

Dr. Rekha B. Venkatapur.

Head of the Deparment ,

K.S.I.T., Bengaluru

Signature of the Principal

Dr. Dilip Kumar K.

Principal/Director,

K.S.I.T., Bengaluru

Internship Completion Certificate

It is certified that **Ms. Rakshita A U D/O A S D Udaya Kumar** Department of **Computer science and Engineering** bearing **USN 1KS22CS113** of K.S. Institute of Technology carried out his/her internship from **2nd Feb 2026** to **Till Date** in this organization **MindMatrix** On the bases of her regularity, punctuality, interest shown towards learning skills, team participation, work experience and meeting internship objectives, a score of Internal Guide and External Guide in total marks out of 100 marks is awarded.

Remarks, if any

Date: **Signature of Mentor:**

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Signature of the College Internship Supervisor

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Signature of the H.O.D **Signature of the Principal**

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Dr. Dilip Kumar K.
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K.S.I.T., Bengaluru

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We would like to express our gratitude to our
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RAKSHITA A U

ABSTRACT

This internship report presents the detailed learning experience and project implementation carried out during the Android App Development using Generative AI internship at MindMatrix. The internship focused on Android development using Kotlin, Android Studio, Room Database, analytics visualization tools, UI/UX principles, and Generative AI integration concepts.

The internship included learning modules such as Artificial Intelligence basics, Prompt Engineering, Data Fundamentals for AI, Android application workflows, Kotlin programming, state management, scalable UI design, analytics implementation, and project-based learning methodologies.

As part of the internship, a project titled "Hasta-Kala Shop" was developed. The application was designed as an Android-based e-commerce platform for artisans and small-scale craft sellers. The application helps users browse handcrafted products, manage orders, track deliveries, and provides an admin dashboard for managing inventory and sales analytics.

The project included features such as Splash Screen, Onboarding, Login/Sign Up, Home Screen, Product Listing, Product Details, Cart, Checkout, Order Tracking, Profile, Wishlist, and Admin Dashboard. Technologies such as Kotlin, Android Studio, Firebase Firestore, Firebase Storage, and XML layouts were explored during project implementation.

The internship improved technical understanding, practical implementation skills, debugging abilities, project management confidence, and communication skills. Overall, the internship provided valuable industry exposure and strengthened readiness for future opportunities in Android development and AI-powered applications.

Keywords: Android Development, Generative AI, Kotlin, Firebase, UI/UX Design, Hasta-Kala Shop, Internship Experience, Industry Readiness, Project Implementation.

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Chapter 1
ABOUT THE COMPANY

1.1 MindMatrix

MindMatrix is a technology company that provides practical training in Android Development, Artificial Intelligence, Machine Learning, and Cloud Computing through industry-oriented internships. The internship environment encouraged innovation, teamwork, and hands-on learning through real-time projects and milestone-based activities. It helped improve technical knowledge, problem-solving abilities, debugging skills, and understanding of Android development workflows, preparing students for future careers in software development and AI technologies.

1.2 Vision and Mission

The vision of MindMatrix is to create a learning ecosystem where quality education is accessible, practical, and aligned with the needs of the modern technology industry. The organization aims to bridge the gap between academic learning and industry requirements by leveraging cutting-edge technologies and innovative mentorship approaches.

The mission of MindMatrix is to equip students with practical, real-world skills that are aligned with current industry demands. The organization focuses on delivering hands-on training, project-based learning, and expert mentorship to ensure that learners gain not only theoretical knowledge but also practical implementation experience. By providing industry-relevant programs, MindMatrix aims to reduce the gap between academic learning and employment requirements.

1.3 Internship Programs

MindMatrix offers a comprehensive range of internship programs focused on modern technology domains. The programs cover areas such as Android Application Development, Artificial Intelligence and Generative AI, Machine Learning, Full Stack Development, and Cloud Computing. These programs are designed with a focus on hands-on learning, real-time projects, and problem-solving skills, which are crucial for technical career roles.

The internship provided over 100 hours of in-depth training covering core and advanced topics such as AI, Android Development, and UI/UX Design. Students received guidance from experienced mentors who offered personalized feedback and industry insights throughout the learning journey.

1.4 Technologies and Tools

During the internship at MindMatrix, various modern technologies and tools were explored. The primary technologies included Kotlin programming language, Android Studio IDE, Firebase Firestore for database management, Firebase Storage for media handling, XML layouts for UI design, and MPAndroidChart for analytics visualization.

Additionally, Generative AI concepts and Prompt Engineering techniques were explored to understand how AI can be integrated into Android applications to provide intelligent features and recommendations. These tools and technologies formed the foundation for the project development activities carried out during the internship.

1.5 Work Culture and Learning Environment

The work culture at MindMatrix was collaborative, innovative, and learning-focused. The internship environment encouraged students to actively participate in project discussions, testing activities, user interface improvements, analytics implementation, and AI-assisted workflow design.

The learning environment promoted continuous improvement and experimentation. Concepts were explained using practical examples, implementation exercises, and milestone-based development activities. Regular feedback sessions, code reviews, and project demonstrations helped maintain high standards of work quality and professional communication throughout the internship period.

Chapter 2
OBJECTIVES OF INTERNSHIP

2.1 Technical Skill Development

One of the primary objectives of the internship was to strengthen technical expertise in Android development, Kotlin programming, and Generative AI integration. The internship provided exposure to modern tools, technologies, and frameworks, enabling practical proficiency and current industry knowledge. Working with technologies such as Kotlin, Firebase, Android Studio, and Generative AI tools helped build a strong technical foundation.

Through hands-on project implementation, the internship facilitated the development of end-to-end Android application development skills, from UI design using XML layouts to backend integration using Firebase Firestore and Storage services.

2.2 To Develop Technical Skills

The internship aimed to bridge the gap between theoretical learning and real-world application. Through this program, students had the opportunity to implement academic concepts into practical scenarios. It helped in understanding how theories are applied in solving real industry problems, thereby improving overall learning effectiveness.

By working on the Hasta-Kala Shop application, practical experience was gained in designing and implementing features such as user authentication, product listing, shopping cart management, order processing, and admin dashboard analytics.

2.3 Industry Exposure

The internship provided exposure to the professional work culture and corporate environment of the technology industry. It helped in understanding how organizations function, including their project workflows, team collaboration practices, communication standards, and delivery timelines.

Students were exposed to real-world project requirements, user experience considerations, and technical constraints that influence software design decisions. This exposure significantly improved the ability to think like a professional developer and approach problem-solving from a product perspective.

2.4 Professional Skill Enhancement

The internship also encouraged the development of essential soft skills such as communication, teamwork, adaptability, time management, and leadership. These skills are crucial for working effectively in a professional environment and interacting with colleagues, mentors, and industry professionals.

Regular project presentations, progress reviews, and documentation activities helped improve the ability to communicate technical concepts clearly and professionally. The internship fostered a growth mindset, encouraging continuous learning and improvement throughout the program.

2.5 Problem Solving

By working on real-time tasks and projects, the internship developed strong problem-solving and logical thinking skills. Students learned how to break down complex problems into smaller, manageable components and approach them in a structured manner. Several technical challenges were encountered during the Hasta-Kala Shop project development, including layout rendering issues, Firebase integration complexities, and state management challenges.

These challenges were resolved through systematic debugging, research, experimentation, and guidance from mentors. This process improved the ability to analyze situations critically, experiment with alternative approaches, and make informed technical decisions.

2.6 Career Readiness

The internship prepared students to become industry-ready professionals by providing practical exposure, skill development, and confidence. It enhanced the professional portfolio, improved technical interview preparedness, and increased chances of securing better job opportunities in Android development and AI-related fields.

The internship experience also provided clarity about career goals and motivated continuous learning and improvement in the field of technology. The combination of technical skills, project experience, and professional development made the internship a significant step toward career readiness.

Chapter 3

LEARNING EXPERIENCE

3.1 AI Fundamentals

The internship began with foundational modules on Artificial Intelligence, covering core concepts such as machine learning, neural networks, natural language processing, and computer vision. Understanding these fundamentals provided the theoretical basis for exploring how AI can be integrated into Android applications to create intelligent, user-centered experiences.

The AI fundamentals modules helped in developing an appreciation for data-driven decision making, model evaluation techniques, and the ethical considerations involved in deploying AI systems. These concepts directly informed the design decisions made during the Hasta-Kala Shop project development.

3.2 Prompt Engineering

Prompt Engineering was a significant learning module during the internship, providing insights into how to effectively communicate with large language models and generative AI systems to achieve desired outputs. The module covered techniques such as zero-shot prompting, few-shot prompting, chain-of-thought reasoning, and role-based prompting.

Practical exercises in prompt engineering improved the ability to leverage AI tools for code generation, documentation writing, debugging assistance, and feature ideation. These skills proved valuable during the project development phase, where AI-assisted tools were used to accelerate development workflows.

3.3 Android Development using Kotlin

The core technical learning during the internship centered around Android application development using the Kotlin programming language. Kotlin's modern features such as null safety, extension functions, coroutines for asynchronous programming, and data classes were explored through hands-on implementation activities.

Key Android development concepts covered included Activity and Fragment lifecycle management, ViewModel and LiveData for state management, RecyclerView for displaying dynamic lists, Navigation Component for screen transitions, and Retrofit for API communication. These concepts were directly applied during the development of the Hasta-Kala Shop application.

3.4 UI/UX Design

UI/UX design principles were explored with a focus on creating intuitive, visually appealing, and accessible Android user interfaces. The module covered material design guidelines, layout design using ConstraintLayout and LinearLayout, custom view creation, animation implementation, and accessibility considerations.

The Hasta-Kala Shop application required careful attention to UI/UX design to create an experience suitable for artisans and craft buyers. The design incorporated warm color palettes reflecting Indian handicraft aesthetics, clear product imagery presentation, and straightforward navigation flows to ensure usability across different user demographics.

3.5 Hasta- Kala Shop Project

The primary project developed during the internship was "Hasta-Kala Shop" – an Android e-commerce application designed specifically for Indian artisans and handcraft sellers. The application enables artisans to showcase their products and allows customers to discover, browse, and purchase unique handcrafted items.

The application was built using Kotlin and Android Studio, with Firebase Firestore serving as the backend database for real-time data synchronization. Firebase Storage was used for product image management. The app featured a comprehensive set of screens including a Splash Screen, Onboarding walkthrough, Login/Sign Up with Google and Phone authentication, Home Screen with categories and best sellers, Product Listing with filtering, Product Details, Shopping Cart, Checkout with multiple payment options, Order Confirmation and Tracking, User Profile management, Wishlist, and an Admin Dashboard for inventory and sales management.

The System Architecture followed a client-server model with the Android app communicating with Firebase services through a defined API layer. The App Flow was designed to guide users smoothly from product discovery through to purchase completion.

The Database Structure in Firestore included collections for Users, Products, Categories, Orders, and OrderItems with appropriate relational references between documents.

3.6 Challenges Faced

During the internship, several technical and organizational challenges were encountered. On the technical side, implementing real-time data synchronization between the Android app and Firebase Firestore required careful handling of asynchronous operations and offline data caching strategies.

UI layout inconsistencies across different screen sizes and Android API levels required extensive testing and adaptation of constraint-based layouts. Integrating multiple Firebase services including Authentication, Firestore, and Storage while maintaining clean architecture principles was complex and required systematic approach and mentor guidance. Debugging navigation state issues and managing the back stack across multiple fragments also presented learning challenges that were overcome through research and experimentation.

Chapter 4

LEARNING OUTCOMES

4.1 Technical Knowledge

The internship significantly enhanced technical knowledge in Android application development, Kotlin programming, Firebase integration, and Generative AI concepts. Hands-on project experience translated theoretical academic knowledge into practical implementation skills applicable in real-world software development scenarios.

Proficiency was gained in working with Kotlin coroutines for asynchronous operations, Firebase Firestore for NoSQL database management, RecyclerView with custom adapters for dynamic content display, and Navigation Component for multi-screen app flows. These technical skills form a strong foundation for professional Android development work.

4.2 Project Experience

Working on the complete lifecycle of the Hasta-Kala Shop application – from requirement analysis and UI/UX design through to implementation, testing, and documentation – provided comprehensive end-to-end project experience. This experience demonstrated how large projects are structured, managed, and delivered in professional settings.

The project experience also highlighted the importance of iterative development, continuous testing, and responsive design in creating robust Android applications. Managing multiple interconnected features such as user authentication, product catalog management, order processing, and analytics dashboards demonstrated the complexity of real-world software projects.

4.3 Analytical Thinking

Working with data throughout the project – from structuring Firestore database schemas to implementing sales analytics in the admin dashboard – developed strong analytical and reasoning skills. The ability to identify patterns, design efficient data structures, and interpret results for business decision-making was significantly improved. Analyzing user experience flows, identifying usability issues, and making data-driven decisions about feature prioritization and design changes strengthened critical thinking and problem analysis capabilities that are essential for professional software development roles.

4.4 Communication Skills

The internship improved communication skills across multiple dimensions. Technical communication was enhanced through documentation writing, code commenting, and project reporting. The internship report itself represents a significant improvement in the ability to organize, present, and communicate complex technical information clearly.

Interpersonal communication skills were developed through regular interactions with the industry mentor, participation in project reviews, and collaborative problem-solving sessions. The ability to ask effective questions, provide project status updates, and discuss technical challenges professionally were all skills that improved throughout the internship period.

4.5 Professional Growth

The internship contributed significantly to professional growth by providing exposure to industry standards, development best practices, and professional work culture. Understanding how technology companies operate, how projects are planned and executed, and how teams collaborate improved overall professional competency.

Completing the Hasta-Kala Shop project from inception to documentation demonstrated the ability to take initiative, manage time effectively, and deliver results independently. This experience boosted confidence in professional settings and provided a concrete example of project delivery capability for future career opportunities.

4.6 Career Readiness

The combination of technical skills, project experience, and professional development gained during the internship significantly improved career readiness. The Hasta-Kala Shop project serves as a strong portfolio piece demonstrating practical Android development capabilities to potential employers.

Understanding industry expectations, development workflows, and professional standards has created a clearer picture of career goals and the preparation required for success in the Android development and AI technology sectors. The internship provided both the technical skills and professional confidence needed to pursue opportunities in these fields effectively.

Chapter 5

CONCLUSION

The internship at MindMatrix has been a highly enriching and valuable learning experience that significantly contributed to academic and professional growth. It provided an opportunity to bridge the gap between theoretical knowledge and practical implementation by working on the Hasta-Kala Shop Android application – a comprehensive e-commerce platform designed for Indian artisans and handcraft sellers.

Through this project, hands-on experience was gained in Android application development using Kotlin, Firebase integration for real-time database and storage management, UI/UX design following material design principles, and Generative AI concepts through prompt engineering modules. The complete development cycle from design to implementation and documentation provided invaluable end-to-end project experience.

The internship also played a crucial role in enhancing soft skills such as communication, teamwork, time management, and adaptability. Overcoming technical challenges such as Firebase synchronization issues, UI layout inconsistencies, and navigation state management helped build problem-solving skills, patience, and confidence.

The Hasta-Kala Shop application successfully demonstrated how modern Android development technologies combined with Firebase backend services can create a fully functional e-commerce solution. The application's features – spanning from user onboarding through product browsing, cart management, checkout, and order tracking to admin dashboard analytics – represent a comprehensive implementation of real-world e-commerce requirements.

Furthermore, the internship has made a significant contribution toward being industry-ready by improving critical thinking, independent work capability, and adaptability to new technologies. It has provided clarity about career goals in Android development and AI-powered applications and motivated continued learning and skill improvement.

In conclusion, the internship has been a significant milestone in the learning journey, providing the technical foundation, practical experience, and professional confidence needed to pursue a successful career in software engineering and Android application development.

Chapter 6

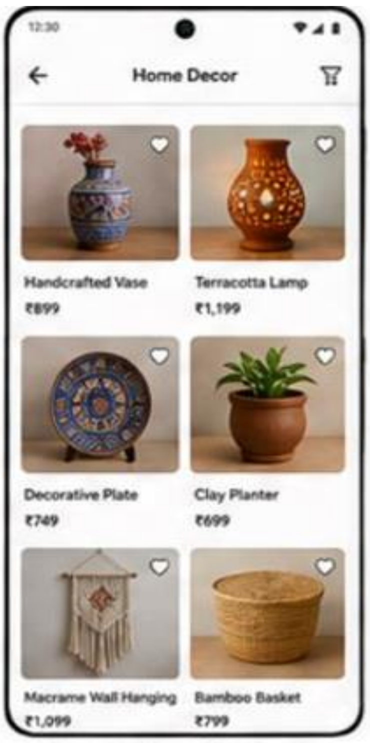
ATTACHMENTS

Fig no. 6.1

Fig no. 6.2

Fig no. 6.3

Fig no. 6.4



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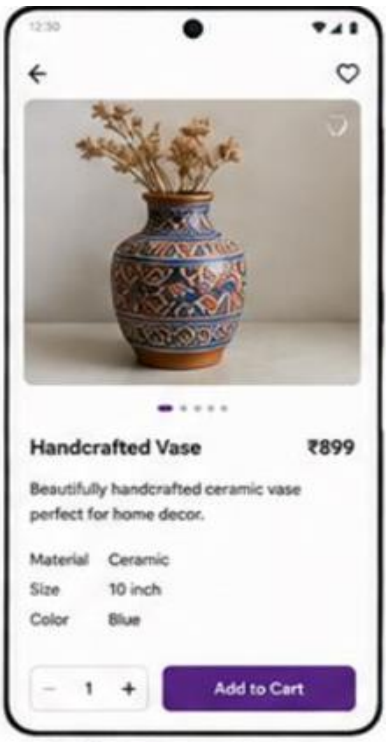

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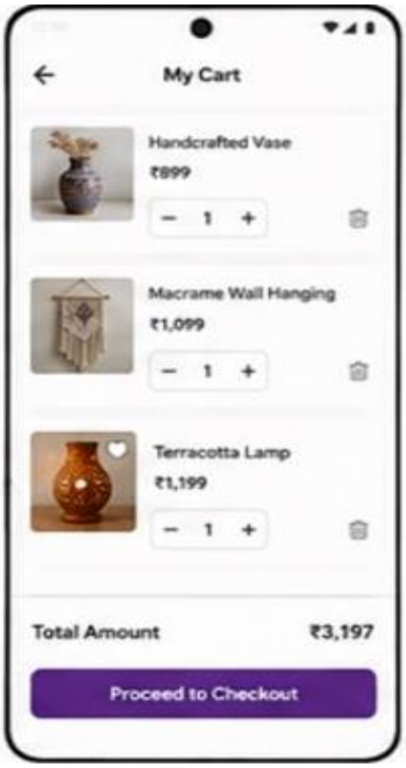

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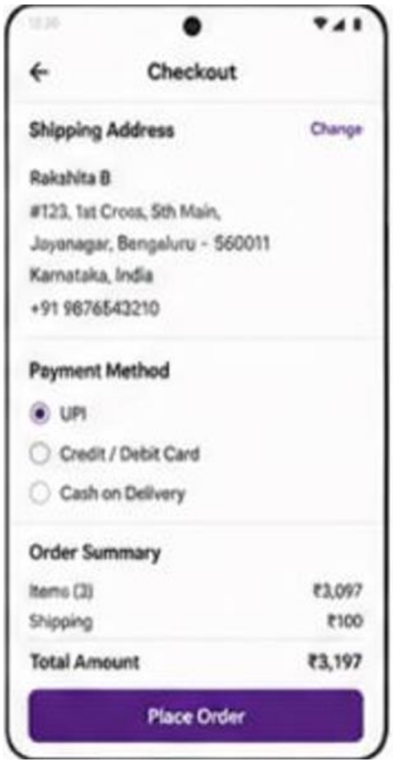

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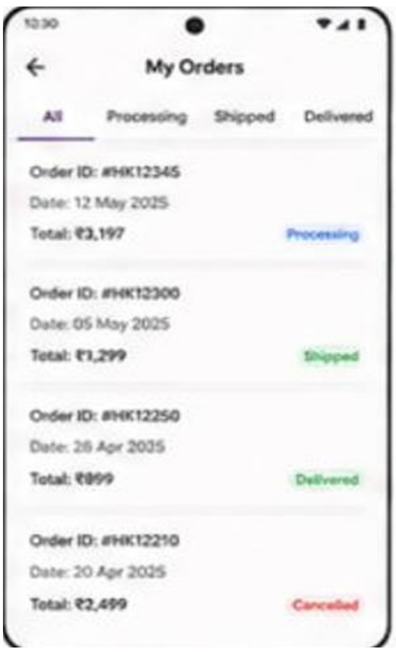

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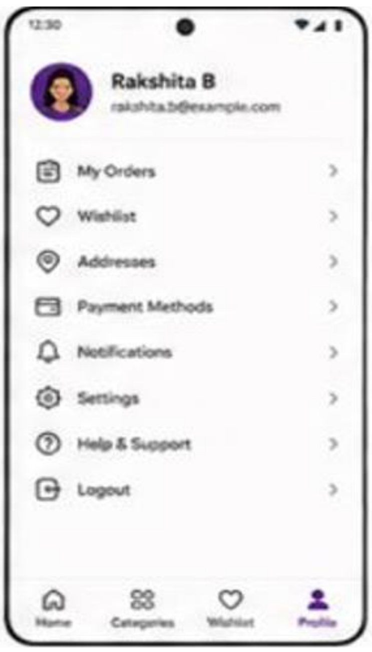

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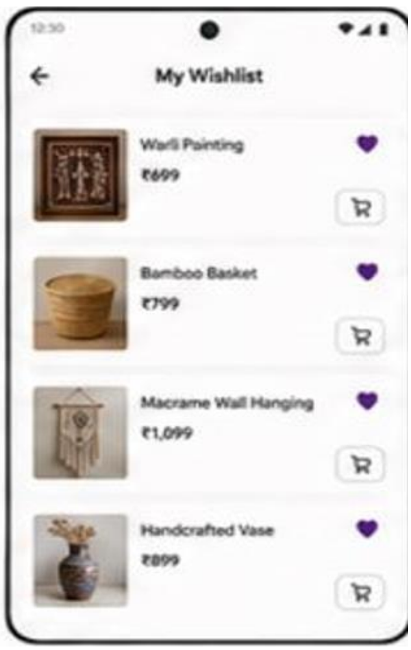

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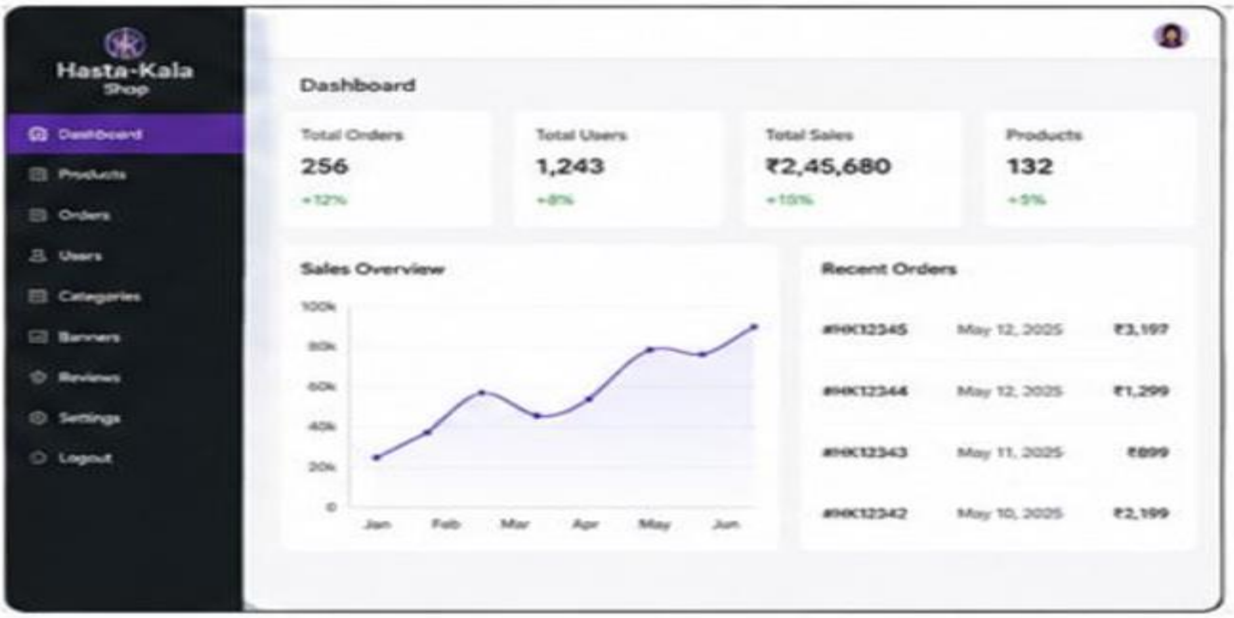

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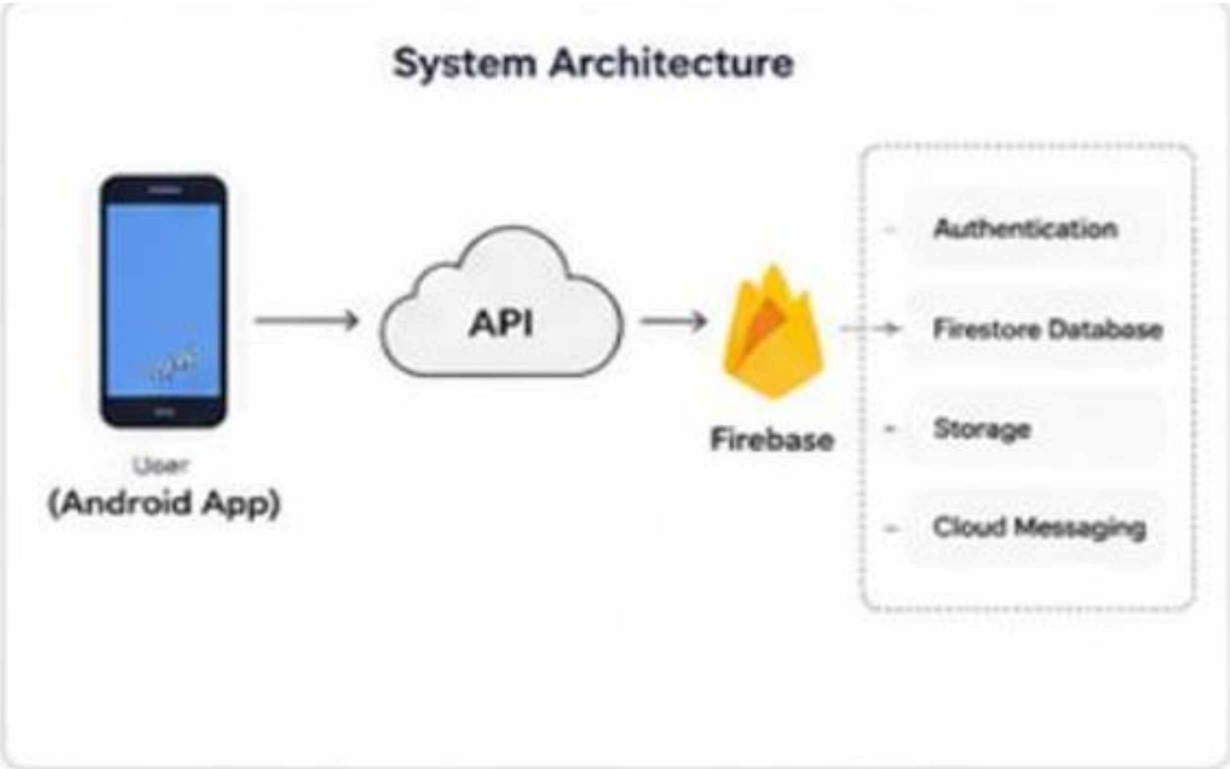

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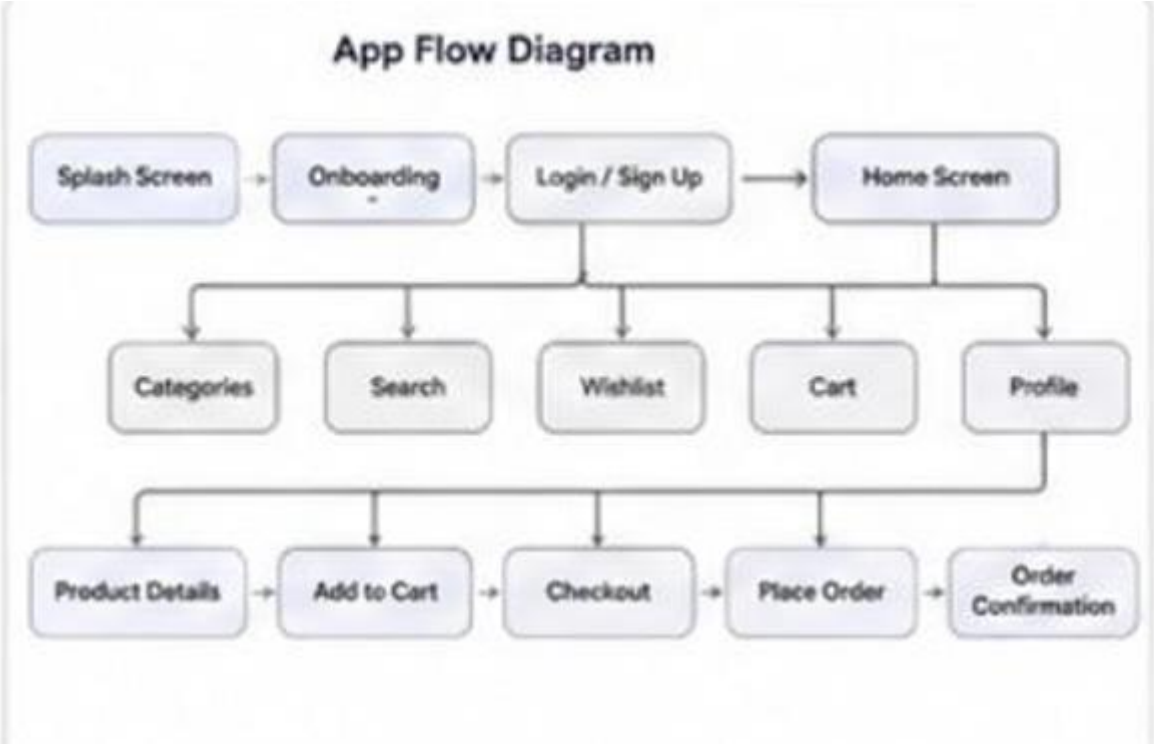

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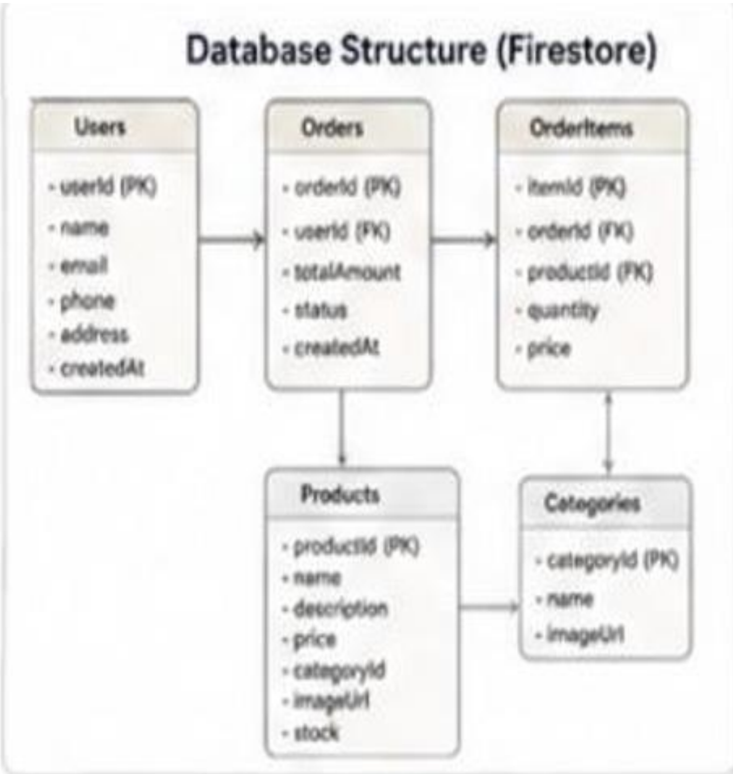

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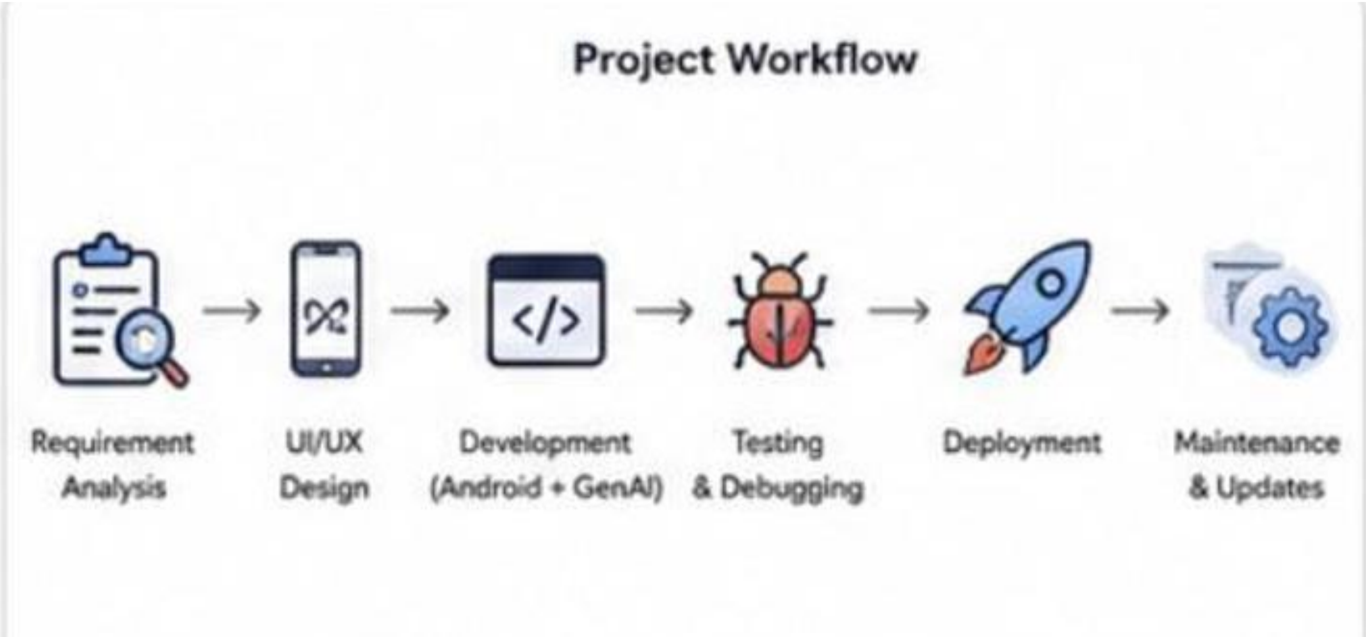
Project Workflow
Requirement Analysis
UI/UX Design
Development (Android + GenAI)
Testing & Debugging
Deployment
Maintenance & Updates

Fig no. 6.17

Chapter 7

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